Urban and Transportation planning lab.
Hiroshima University

CSIS DAYS 2026

Beyond the Building: A Probabilistic Framework for GPS Assignment Under Positional Uncertainty

Hafiyyan Hilmy Fawwaz

Supervisor: Prof. Makoto Chikaraishi

Graduate School of Innovation and Practice for Smart Society · Hiroshima University

where did they really stop?

- A phone records a **position**, not a **place**.
- In a city, GPS error is often bigger than the gap between a building, the street in front of it and the open space next to it.
- The usual fix is to pick the **nearest building**.
- But daily life also happens outside buildings.[1]
- And one answer hides how unsure the GPS is.



Building Street Open space

7–13 m

typical smartphone GPS error in a city[2]

1. Jacobs, 1961; Gehl, 2010; Whyte, 2010 2. Merry & Bettinger, 2019

the idea: give every nearby place a probability

1 · Find candidates
buildings, street pieces, open spaces within 10 m

→

2 · Give four scores
for each candidate

→

3 · Add them up
each score has a weight

→

4 · Make probabilities
a logit model, one per place type

Time near

share of the stay spent close to it

Distance

how far it is from the stay centre

Longest stretch

longest unbroken time at it

Direction

movement along the street

> The weights are not set by hand. They are learned from the answers of a field survey.

a field survey with known answers

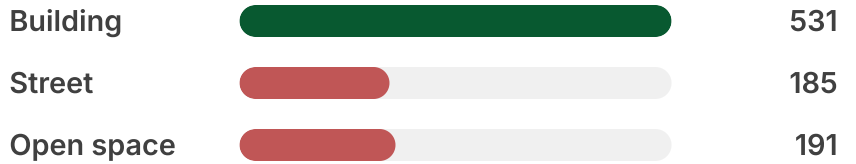
11
people, 2 sessions each

3–5
phones per person

280
real stops

907
stays with a known answer

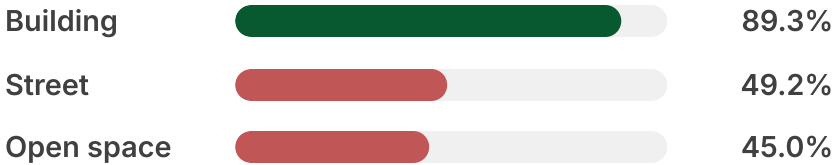
People walked around Hiroshima in June 2026. At each planned stop they wrote down the **place type**, and when they arrived and left.



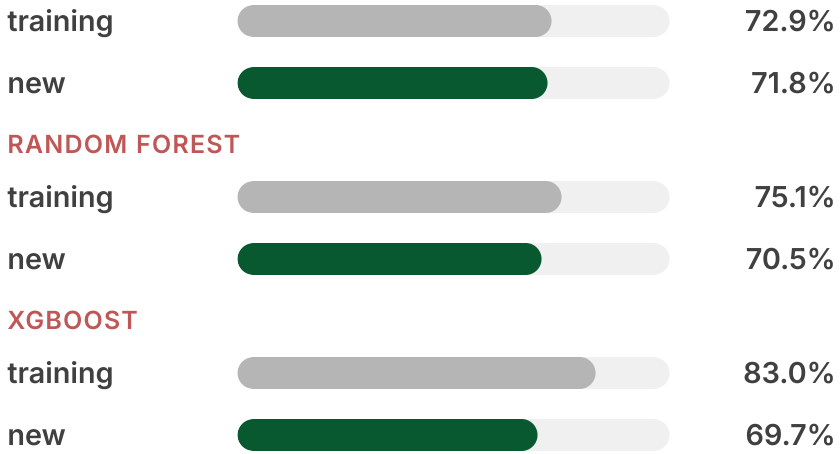
right 7 times in 10, even on new stays

71.8%

of new stays get the right place type. "New" means the model did not learn from them.



Is a bigger model better? No.



The big models fit the training stays better. On new stays, our small model is the best.

three takeaways

New

A probability for building, street and open space, for people who stay in one place.

Simple is enough

11 numbers beat two machine-learning models on new stays.

Fixable

For streets, the weak point is how the map cuts them into pieces. It is not the GPS data.

The slides after this one are the **attachment**: the method with all formulas, the data, and every result. Contact: hafifawwaz@hiroshima-u.ac.jp

attachment: the full story

For readers who scan the poster. The method follows the master's thesis, and formulas are numbered in order of appearance.

1

Background

2

Data

3

Method

4

Results

5

Limits and next

the gap in the literature

Deterministic methods

Alvares et al. 2007 · Wu et al. 2026

A stay goes to the building or point of interest that contains it or lies nearest. One answer, no confidence. Wu et al. give a probability only as a check after the fact.

Probabilistic, for moving

Bierlaire et al. 2013 · Chen & Bierlaire 2015 · Kempinska et al. 2016

Each position depends on a neighbouring point or the previous time step. A single stay has no such chain.

This study

stays, not routes

Keeps the probabilistic idea. The scores come from the properties of one static cluster, and the targets are buildings, streets and open spaces.

Streets and open spaces matter: Jacobs (1961), Gehl (2010) and Whyte (2010) place everyday public life there, and Sherwani et al. (2025) counted 3,565 outdoor social activities in four neighbourhoods. A few metres of GPS error can move a point from one entity to another (Weng et al., 2025).

three research questions

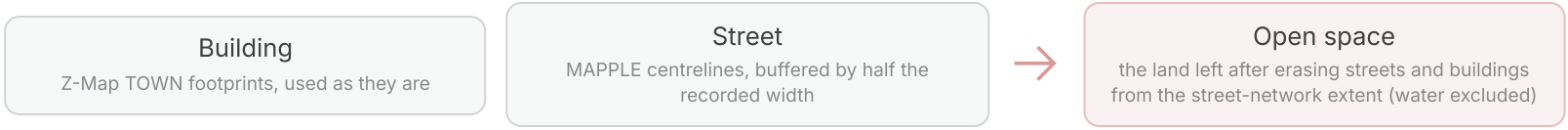
RQ1 How can a scoring framework include positional uncertainty and treat buildings, streets and open spaces as valid targets, without fixing the weight of each evidence source in advance?

RQ2 How close are the assignments to independently observed ground truth, and what drives the mismatches?

RQ3 Does added model complexity improve the fit, or only tighten it to the calibration sample?

study area and the three entity layers

Hiroshima City. Positions use one coordinate system (WGS 84 / UTM zone 53N, EPSG:32653).



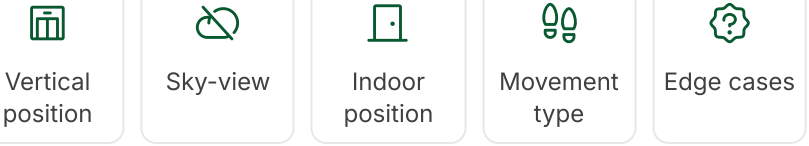
Why footprints, not Voronoi. Voronoi regions cover everything, but a large region can capture points that did not belong to its building. Footprints are more precise, so open space is the residual instead.

A known limit. Z-Map TOWN dates from 2020 and the survey from 2026. A structure met in the field may have no polygon, most clearly underground shopping arcades.

the field experiment

Participants	11, two sessions each
Session length	about 8 hours
Stops per session	12 controlled + 4 free-roam
Devices per person	3 (4 or 5 in six sessions)
Logging	highest accuracy, 1.0 m minimum distance
Sessions	two batches in June 2026

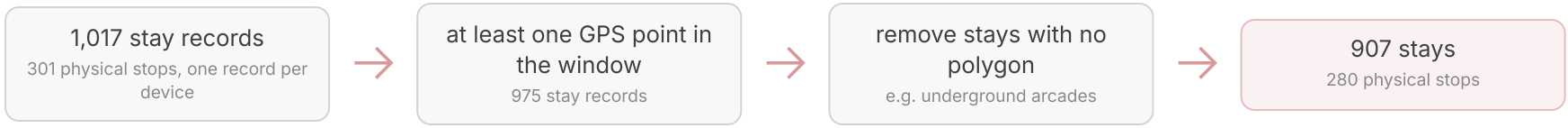
FIVE THEMATIC GROUPS



At each stop, a self-report gives the place type, floor, sky-view and indoor position. The reported start and end times set the stay boundaries.

from reports to 907 stays

GROUND TRUTH



Matching the application data. The target is the median density of MILES stays found with ST-DBSCAN: 0.967 points per minute. Survey points were downsampled at random, keeping each cluster's first and last point: 152,602 points became 15,687.

Limits. The protocol makes movement artificial. Some devices ran out of battery. Up to five devices recorded the same stop, so stays are not independent: cross-validation groups them by stop.

framework: find the candidates

RQ1

Each stay cluster C_i gets a probability for each candidate entity. A candidate is any entity whose boundary lies within the buffer distance of at least one GPS point of the cluster.

$$\mathrm{Cand}(C_i) = \{E_k \mid \min_{p \in C_i} d(p, E_k) \leq R_{\mathrm{buffer}}\} \tag{1}$$

E_k entity · p GPS point · d distance · R_{buffer} buffer, 10 m

Why a buffer. GPS error can push a point outside the true boundary. A buffer replaces strict containment. 10 m beat 15 m on every model tested.

score 1: spatial distribution RQ1

The time-weighted share of the cluster's GPS points within the buffer distance of a candidate.

$$f_{\text{distribution}}(E_k, C_i) = \frac{\sum_{p \in C_i} W_p \, \mathbf{1}\{d(p, E_k) \le R_{\text{buffer}}\}}{\sum_{p \in C_i} W_p}$$

(2)

Time weight. $W_p = (t_{p+1} - t_{p-1})/2$ for interior points, and half the gap to the neighbour at the cluster ends. Timestamps are rescaled to $[0, 1]$, so the weights sum to one.

Intuitively. The score is high when the phone spent most of the stay close to the candidate. It does not assume that GPS samples are evenly spaced.

score 2: distance decay RQ1

Closer candidates score higher. The cluster's own spread sets how fast the score falls.

$$f_{\text{distance}}(E_k, C_i) = \exp\left(-\frac{d_{ki}^2}{2(\alpha \cdot \sigma_i)^2}\right) \tag{3}$$

d_{ki} distance from the cluster medoid to E_k (zero if the medoid is inside) · σ_i spatial standard deviation of the cluster, floored at 5 m · α scale, calibrated once on ground truth: $\alpha^* = 2.979$.

Why dispersion. A spread-out cluster is a less certain location. Map matching also uses GPS variance for confidence regions (Quddus et al., 2007), and cluster extent helps stay location estimates (Iwata et al., 2018).

score 3: temporal concentration RQ1

The longest unbroken, time-weighted run of points that belong to the candidate.

$$f_{\text{temporal}}(E_k, C_i) = \frac{\max_S \sum_{p \in S} W_p}{\sum_{p \in C_i} W_p} \tag{4}$$

S is a run of points within R_{buffer} of E_k. A point outside the buffer neither breaks nor ends a run if it is recorded within $\Delta t_{\text{gap}} = 300$ s of the run's last point.

Bridging. A run breaks only at an observed point outside the buffer, recorded more than 300 s after the last point of the run. A stretch with no GPS points never breaks a run.

score 4: directional alignment

RQ1

How well the small movements inside the cluster follow the axis of street E:

$$A(E, C_i) = \frac{1}{|J|} \sum_{j \in J} |\cos(\theta_j - \theta_E)| \tag{5}$$

θ_j bearing between consecutive points · J steps longer than 0.5 m · θ_E length-weighted mean axis of the street centreline.

For a street, the score is the alignment itself. For a building or open space, it is one minus the best alignment with any street within 30 m:

$$f_{\text{directional}}(E_k, C_i) = \begin{cases} A(E_k, C_i) & E_k \text{ a street} \\ 1 - \max\limits_{s \in \text{Adj}(E_k)} A(s, C_i) & \text{otherwise} \end{cases} \tag{6}$$

adding the scores up

RQ1

The four scores become one utility for each candidate, through a weighted sum.

$$V_k = w_1 f_{\text{distribution}} + w_2 f_{\text{distance}} + w_3 f_{\text{temporal}} + w_4 f_{\text{directional}} \tag{7}$$

Why a sum, not a product. In a product, one score near zero would remove a candidate whatever the other three say. A score can be near zero in some places without meaning that no activity happened there.

The weights $w_1, \ldots, w_4$ are not set in advance. Conditional logit estimation on the ground truth finds them (calibration, below).

from utilities to probabilities

RQ1

The result is judged by place type, so the utilities of all candidates of one type are combined with a multinomial logit transform.

$$P(t \mid C_i) = \frac{\sum_{k \in t} \exp(V_k)}{\sum_{j \in \text{Candidates}(C_i)} \exp(V_j)}, \quad t \in \{\text{building}, \text{street}, \text{open space}\}$$

(8)

The exponential keeps every probability above zero, and it matches the multinomial logit of discrete choice models (McFadden, 1973).

The three probabilities add up to one. The predicted type is the one with the highest probability.

calibration: step by step

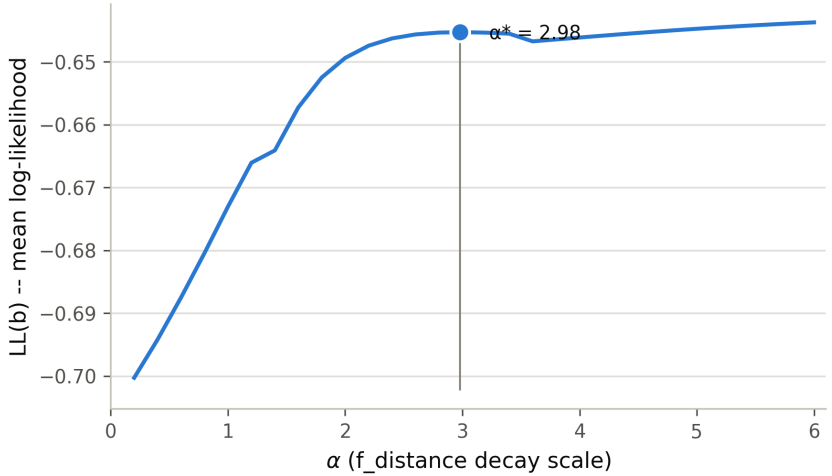
RQ1

Models were fitted in sequence. Each step asks whether the fit gain is worth its reliability cost. Accuracy is cross-validated with folds grouped by physical stop.

Model	Params	In-sample	CV accuracy	Cond. no.	Why
base	4	0.519	0.522 ± 0.055	86.0	generic weights only
+ share + σ	7	0.663	0.655 ± 0.029	114.3	drops f_{temporal} (collinear)
alternative-specific	13	0.714	0.697 ± 0.026	739.3	a weight per type
+ ASC (rejected)	15	0.728	0.719 ± 0.016	2,055.5	duplicates the street direction score (VIF 31.0)
+ interactions	16	0.716	0.709 ± 0.037	13,815.0	no gain in conditioning
calibrated (α = 2.979)	13	0.727	0.710 ± 0.030	2,036.4	α set once, reused
final	**11**	**0.729**	**0.718 ± 0.028**	**1,868.6**	one shared direction weight

calibrating the distance scale α

RQ1



profile likelihood for α

- The best value inside the range is $\alpha^* = 2.979$.
- The curve is flat above about 2.4: every α from there upward is within the 95% likelihood-ratio limit.
- Above about 4 the score tends to 1 for almost every candidate and acts like a per-type constant, the rejected ASC in another form. So the interior optimum is used.

the final model RQ1

11 parameters, $\alpha = 2.979$. The direction score is shared across types: a separate weight per type is not justified (LR test, $p = 0.115$).

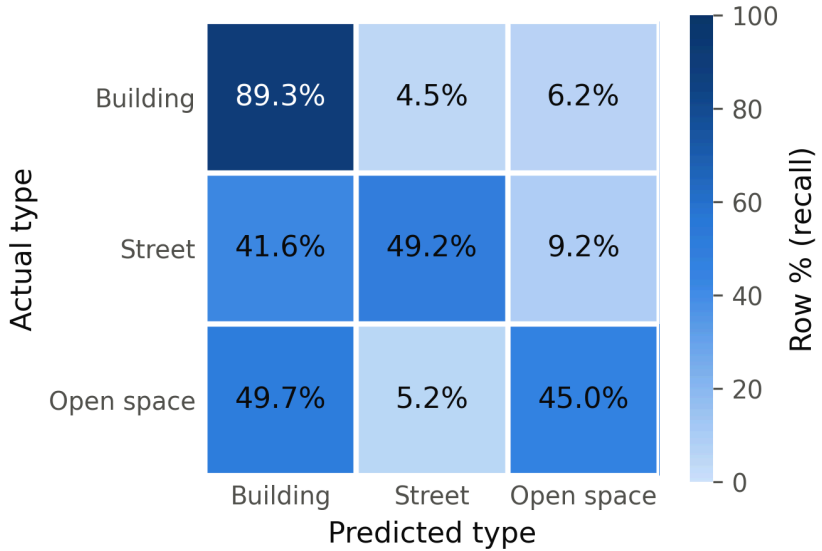
$$V_t = w_{1,t} f_{\text{distribution}} + w_{2,t} f_{\text{distance}} + w_3 f_{\text{directional}} + w_{4,t}\,\text{share}_t + w_5\,\sigma \tag{9}$$

Weight (t-value)	Building	Street	Open space
$w_{1,t} \cdot f_{\text{distribution}}$	+1.960 (4.06)	+2.659 (5.36)	+6.410 (11.01)
$w_{2,t} \cdot f_{\text{distance}}$	−0.465 (−0.37)	−3.350 (−2.92)	−6.665 (−5.16)
$w_{4,t} \cdot \text{share}_t$	−1.182 (−2.38)	−5.156 (−7.27)	+0.246 (0.23)

Shared across types: $w_3 = +3.488$ ($f_{\text{directional}}$, t = 6.82) and $w_5 = +1.045$ (σ, t = 7.63). share_t is the fraction of a stay's candidates that are of type t. Cross-validated accuracy is **0.718 ± 0.028**, condition number 1,869.

validation: where the errors go

RQ2

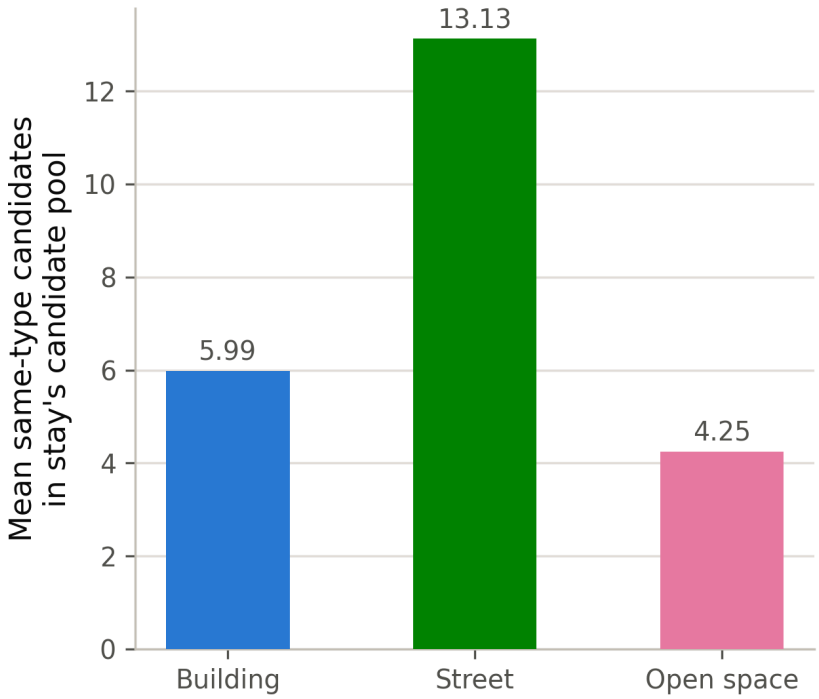


Out-of-sample accuracy is **0.718**.

- Building is found 89.3% of the time (474 of 531). Street 49.2% (91 of 185). Open space 45.0% (86 of 191).
- A missed street stay goes to building four and a half times as often as to open space.
- A missed open space stay goes to building almost ten times as often as to street.
- More open space stays are predicted as building (95) than correctly (86).

why streets are missed

RQ2



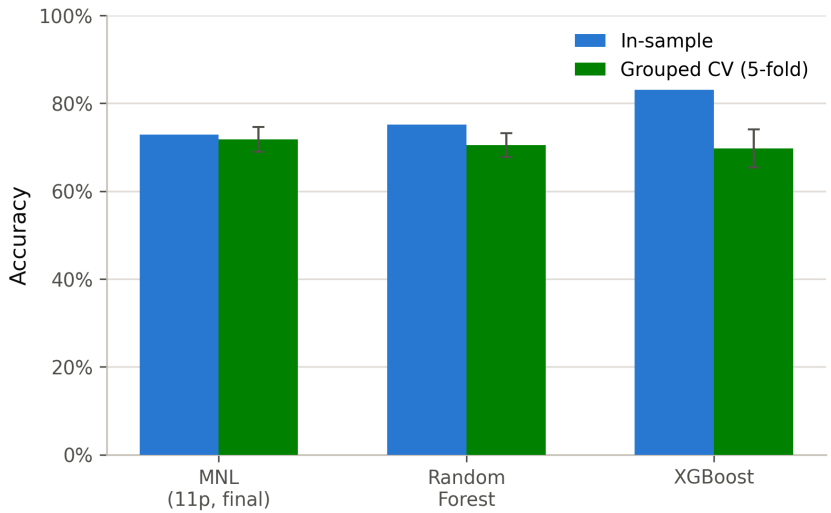
same-type candidates in a true stay's candidate pool

- The share_t weight punishes a candidate for each same-type competitor around it. The street weight is the largest: −5.16, against −1.18 for building.
- A street stay has 13.1 street candidates nearby, a building stay 6.0 building candidates, an open space stay 4.2.
- The map cuts one street into many short pieces, so streets are punished before the GPS evidence counts.

> Open space has no such explanation. Its errors are still open.

a bigger model does not help

RQ3



Random forest and XGBoost use the same inputs and the same folds.

Model	In-sample	Grouped CV
MNL (final)	0.729	**0.718 ± 0.028**
Random forest	0.751	0.705 ± 0.027
XGBoost	0.830	0.697 ± 0.044

- The gap between training and new stays: MNL −1.1, random forest −4.6, XGBoost −13.4 points.
- The shortfall sits in open space: 0.450 for the MNL, 0.282 and 0.345 for the ensembles.
- Refitting on raw, non-downsampled points gives a slightly lower CV accuracy (0.709).

limits and what comes next

Limits

- The ground truth was collected under a controlled protocol, and some stays were dropped.
- Up to five devices share one stop, so the data are not fully independent.
- Street results describe this map representation, not the method itself.
- The open space errors are not explained yet.
- Each stay is assigned alone, without the stays before it.

Next

- **Other cities and MILES.** The type mix of the survey (59% building, 20% street, 21% open space) is by design, so transfer to city-scale data is open.
- **Street segmentation.** Test fewer, longer street pieces.
- **Sequence-aware assignment.** Use the stays before and after as context.

references

Alvares, L. O., et al. (2007). A model for enriching trajectories with semantic geographical information. *ACM GIS*.

Bierlaire, M., et al. (2013). A probabilistic map matching method for smartphone GPS data. *Transp. Res. C*, 26.

Birant, D. & Kut, A. (2007). ST-DBSCAN. *Data & Knowl. Eng.*, 60.

Chen, J. & Bierlaire, M. (2015). Probabilistic multimodal map matching with rich smartphone data. *J. Intell. Transp. Syst.*

Gehl, J. (2010). *Cities for People*. Island Press.

Iwata, S., et al. (2018). A stayed location estimation method for sparse GPS positioning information. *IEICE Trans. Fundam.*, E101.A.

Jacobs, J. (1961). *The Death and Life of Great American Cities*.

Kato, R., et al. (2024). Evaluating and validating stay point detection algorithms. *Procedia Comput. Sci.*, 238.

Kempinska, K., et al. (2016). Probabilistic map-matching using particle filters. *arXiv*.

McFadden, D. (1973). Conditional logit analysis of qualitative choice behavior.

Merry, K. & Bettinger, P. (2019). Smartphone GPS accuracy study in an urban environment. *PLOS ONE*.

Quddus, M. A., et al. (2007). Current map-matching algorithms for transport applications. *Transp. Res. C*, 15.

Sherwani, A. B., et al. (2025). Spatiality of outdoor social activities in neighborhood urban spaces.

Weng, D., et al. (2025). Sidewalk matching: a smartphone-based GNSS positioning technique for pedestrians.

Whyte, W. H. (2010). *The Social Life of Small Urban Spaces*.

Wu, H., et al. (2026). An object-oriented spatial statistics approach for human activity space estimation. *arXiv*:2605.08509.

Urban and Transportation planning lab.
Hiroshima University

THANK YOU

Beyond the Building

Hafiyyan Hilmy Fawwaz

hafifawwaz@hiroshima-u.ac.jp

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